

Appendix: Supplementary material data

Table A1: Species list of collected wild bees from forest steppe study sites including information on species functional traits. Functional traits information after Bees, Wasps & Ants Recording Society (BWARS); Bogusch & Straka, 2012; Falk, 2015; Michez et al., 2019.

Taxa	Social habit	Nest location	Larvae feeding preference	Body size (mm)
1. <i>Amegilla quadrifasciata</i> (de Villers, 1789)	Solitary	Soil	polylectic	11
2. <i>Andrena combaella</i> Warncke, 1966	Solitary	Soil	oligolectic	12
3. <i>Andrena niveata</i> Friese, 1887	Solitary	Soil	oligolectic	11.5
4. <i>Andrena oralis</i> Morawitz, 1876	Solitary	Soil	polylectic	12
5. <i>Anthidium interruptum</i> (Fabricius, 1781)	Solitary	Soil	polylectic	12
6. <i>Anthidium manicatum</i> (Linnaeus, 1758)	Solitary	Woody plant	polylectic	11.5
7. <i>Bombus humilis</i> Illiger, 1806	Eusocia l	Soil	polylectic	12
8. <i>Bombus pascuorum</i> (Scopoli, 1763)	Eusocia l	Soil	polylectic	12
9. <i>Bombus ruderatus</i> (Fabricius, 1775)	Eusocia l	Soil	polylectic	12.5
10. <i>Bombus terrestris</i> (Linnaeus, 1758)	Eusocia l	Soil	polylectic	14
11. <i>Coelioxys quadridentata</i> (Linnaeus, 1758)	Parasiti c	Dead wood	polylectic	8
12. <i>Halictus confusus</i> Smith, 1853	Eusocia l	Soil	polylectic	7
13. <i>Halictus semitectus</i> Morawitz, 1874	Solitary	Soil	polylectic	6
14. <i>Hoplitis bisulca</i> (Gerstäcker, 1869)	Solitary	Soil	oligolectic	10
15. <i>Hylaeus angustatus</i> (Schenck, 1861)	Solitary	Dead wood	polylectic	4
16. <i>Hylaeus brevicornis</i> Nylander, 1852	Solitary	Herbaceous plants	polylectic	5
17. <i>Hylaeus communis</i> Nylander, 1852	Solitary	Woody plant	polylectic	6.5
18. <i>Hylaeus gibbus</i> Saunders, 1850	Solitary	Herbaceous plants	polylectic	7
19. <i>Lasioglossum albipes</i> (Fabricius, 1781)	Eusocia l	Soil	polylectic	8
20. <i>Lasioglossum bluethgeni</i> Ebmer, 1971	Solitary	Soil	oligolectic	7
21. <i>Lasioglossum brevicorne</i> (Schenck, 1869)	Eusocia l	Soil	oligolectic	6.5
22. <i>Lasioglossum calceatum</i> (Scopoli, 1763)	Eusocia l	Soil	polylectic	8.5
23. <i>Lasioglossum leucozonium</i> (Schrank, 1781)	Eusocia l	Soil	polylectic	9
24. <i>Lasioglossum mesosclerum</i> (Perez, 1903)	Solitary	Soil	polylectic	4
25. <i>Lasioglossum minutissimum</i> (Kirby, 1802)	Solitary	Soil	polylectic	4.5
26. <i>Lasioglossum politum</i> (Schenck, 1853)	Eusocia l	Soil	polylectic	4.5
27. <i>Lasioglossum punctatissimum</i> (Schenck, 1853)	Solitary	Woody plant	polylectic	6.5
28. <i>Lasioglossum puncticolle</i> (Morawitz, 1872)	Eusocia l	Soil	polylectic	6.5
29. <i>Lasioglossum semilucens</i> (Alfken, 1914)	Solitary	Soil	polylectic	5.5

30. <i>Lasioglossum sexstrigatum</i> (Schenck, 1869)	Solitary	Soil	polylectic	6.5
31. <i>Lasioglossum tarsatum</i> (Schenck, 1870)	Solitary	Soil	polylectic	5.5
32. <i>Lasioglossum xanthopus</i> (Kirby, 1802)	Solitary	Herbaceous plants	polylectic	12
33. <i>Lasioglossum zonulum</i> (Smith, 1848)	Solitary	Soil	polylectic	9.5
34. <i>Megachile centuncularis</i> (Linnaeus, 1758)	Solitary	Dead wood	polylectic	9
35. <i>Megachile circumcincta</i> Kirby, 1802	Solitary	Dead wood	polylectic	10
36. <i>Megachile leachella</i> Curtis, 1828	Solitary	Soil	polylectic	9.5
37. <i>Megachile maritima</i> (Kirby, 1802)	Solitary	Woody plant	polylectic	14.5
38. <i>Megachile melanopyga</i> Costa, 1863	Solitary	Woody plant	oligolectic	11.5
39. <i>Megachile versicolor</i> Smith, 1844	Solitary	Herbaceous plants	polylectic	11.5
40. <i>Nomioides minutissimus</i> (Rossi, 1790)	Solitary	Woody plant	polylectic	4.5
41. <i>Osmia brevicornis</i> (Fabricius, 1798)	Solitary	Dead wood	oligolectic	11
42. <i>Osmia melanogaster</i> Spinola, 1808	Solitary	Dead wood	oligolectic	9
43. <i>Sphecodes puncticeps</i> Thomson, 1870	Parasitic	Soil	polylectic	6
44. <i>Trachusa interrupta</i> (Fabricius, 1781)	Solitary	Soil	polylectic	12
45. <i>Xylocopa iris</i> (Christ, 1791)	Solitary	Dead wood	polylectic	16.5

Table A2: Species list of collected wasps from forest steppe study sites including information on species functional traits. Functional traits information after Agnoli & Rosa, 2021; Bees, Wasps & Ants Recording Society (BWARS); Evans & Matthews, 1973; Witt, 1998; Yeo & Corbet, 2015.

	Taxa	Social habit	Nest location	Wasp preys	Larvae feeding preference	Body size (mm)
1.	<i>Ancistrocerus gazella</i> (Panzer 1798)	Solitary	Herbaceous plants	Caterpillars (Lepidoptera)	herbivore	11
2.	<i>Aporinellus moestus sericeomaculatus</i> (Kohl, 1888)	Solitary	Soil	Spiders (Araneae)	carnivore	4
3.	<i>Arachnospila abnormis</i> (Dahlbom, 1842)	Solitary	Herbaceous plants	Spiders (Araneae)	carnivore	9
4.	<i>Arachnospila minutula</i> (Dahlbom, 1842)	Solitary	Soil	Spiders (Araneae)	carnivore	8
5.	<i>Arachnospila spissa</i> (Schjødte, 1837)	Solitary	Soil	Spiders (Araneae)	carnivore	8
6.	<i>Arachnospila trivialis</i> (Dahlbom, 1843)	Solitary	Soil	Spiders (Araneae)	carnivore	6
7.	<i>Astata boops</i> (Schränk, 1781)	Solitary	Soil	True bugs (Hemiptera)	herbivore	11
8.	<i>Bembecinus tridens</i> (Fabricius, 1781)	Solitary	Soil	True bugs (Hemiptera)	herbivore	10
9.	<i>Chrysis illigeri</i> Wesm., 1839	Parasitic	Soil	Wasps (Hymenoptera: Vespidae)	carnivore	2.5
10.	<i>Chrysis rutilans</i> Olivier, 1790	Parasitic	Herbaceous plants	Wasps (Hymenoptera: Vespidae)	carnivore	8
11.	<i>Chrysura dichroa</i> (Dahlbom, 1854)	Parasitic	Woody plant	Bees (Hymenoptera: Apidae)	herbivore	3
12.	<i>Crabro scutellatus</i> (von Scheven, 1781)	Solitary	Woody plant	Dolichopodidae (Diptera, Aphidae predators)	herbivore	7

13. <i>Crossocerus wesmaeli</i> (Vander Linden, 1829)	Solitary	Soil	Small flies (Diptera)	herbivore	5
14. <i>Diodontus minutus</i> (Fabricius, 1793)	Solitary	Herbaceous plants	Aphids (Aphidoidea)	herbivore	5
15. <i>Dipogon bifasciatus</i> (Geoffroy, 1785)	Solitary	Herbaceous plants	Spiders (Araneae)	carnivore	6
16. <i>Ectemnius dives</i> (Lepelletier de Saint Fargeau & Brullé, 1835)	Solitary	Dead wood	Hoverflies (Syrphidae), true flies (Tachinidae)	herbivore	8
17. <i>Episyron albonotatus</i> (Costa, 1881)	Solitary	Soil	Spiders (Araneae)	carnivore	13
18. <i>Episyron rufipes</i> (Linnaeus 1758)	Solitary	Soil	Spiders (Araneae)	carnivore	11
19. <i>Evagetes pectinipes</i> (Linnaeus, 1758)	Parasitic	Soil	Wasps (Hymenoptera: Vespidae)	carnivore	5
20. <i>Gorytes quinquecinctus</i> (Fabricius, 1793)	Solitary	Soil	Small leafhoppers (Auchenorrhyncha)	herbivore	10
21. <i>Meria tripunctata</i> (Rossi, 1790)	Parasitic	Dead wood	Black beetle (Tenebrionidae)	herbivore	14
22. <i>Nemka viduata</i> (Pallas, 1773)	Parasitic	Herbaceous plants	Wasps (Hymenoptera: Vespidae)	carnivore	16
23. <i>Oxybelus quatordecimnotatus</i> Jurine, 1807	Solitary	Soil	Muscids (Diptera: Muscidae)	herbivore	6
24. <i>Passaloecus gracilis</i> (Curtis, 1834)	Solitary	Herbaceous plants	Aphids (Aphidoidea)	herbivore	4
25. <i>Passaloecus pictus</i> Ribaut, 1952	Solitary	Soil	Aphids (Aphidoidea)	herbivore	6
26. <i>Pemphredon inornata</i> (Say 1824)	Solitary	Woody plant	Aphids (Aphidoidea)	herbivore	1.5
27. <i>Pemphredon lethifera</i> (Shuckard, 1837)	Solitary	Woody plant	Aphids (Aphidoidea)	herbivore	1.5
28. <i>Pemphredon lugens</i> (Dahlbom 1842)	Solitary	Woody plant	Aphids (Aphidoidea)	herbivore	1.5
29. <i>Philanthus venustus</i> (Rossi, 1790)	Solitary	Soil	Bees (Hymenoptera: Apidae)	herbivore	2
30. <i>Polistes nimpha</i> (Christ, 1791)	Eusocial	Herbaceous plants	Spiders (Araneae)	carnivore	8
31. <i>Pompilus cinereus</i> Fabricius, 1798	Solitary	Soil	Spiders (Araneae)	carnivore	6
32. <i>Pseudomalus auratus</i> (Linnaeus, 1758)	Parasitic	Woody plant	Wasps (Hymenoptera: Vespidae)	carnivore	5
33. <i>Pseudomalus pusillus</i> (Fabricius, 1804)	Parasitic	Herbaceous plants	Wasps (Hymenoptera: Vespidae)	carnivore	4
34. <i>Scolia hirta</i> (Schrank, 1781)	Solitary	Dead wood	Scarabaeidae	herbivore	24
35. <i>Sphex funerarius</i> Gussakovskij, 1934	Solitary	Soil	Grasshoppers, crickets (Orthoptera)	herbivore	20
36. <i>Stenodynerus chevrieranus</i> (Saussure, 1855)	Solitary	Soil	Other insects	omnivore	7
37. <i>Symmorphus gracilis</i> (Brullé, 1832)	Solitary	Herbaceous plants	Leaf beetle (Chrysomelidae)	herbivore	10
38. <i>Tachysphex mediterraneus</i> Kohl, 1883	Solitary	Soil	Grasshoppers, crickets (Orthoptera)	herbivore	8

39. <i>Tachysphex obscuripennis</i> (Schenck, 1857)	Solitary	Soil	Grasshoppers, crickets (Orthoptera)	herbivore	8
40. <i>Tachysphex pompiliformis</i> (Panzer, 1805)	Solitary	Soil	Grasshoppers, crickets (Orthoptera)	herbivore	6
41. <i>Tachysphex psammobius</i> (Kohl, 1880)	Solitary	Soil	Grasshoppers, crickets (Orthoptera)	herbivore	5
42. <i>Tachysphex unicolor</i> (Panzer, 1809)	Solitary	Soil	Acrididae	herbivore	6
43. <i>Telostegus inermis</i> (Brullé, 1832)	Solitary	Soil	Spiders (Araneae)	carnivore	7
44. <i>Tiphia femorata</i> Fabricius, 1775	Parasitic	Soil	Scarabs (Scarabaeidae: Amphimallon)	herbivore	10
45. <i>Trypoxylon scutatum</i> Chevrier 1867	Solitary	Herbaceous plants	Spiders (Araneae)	carnivore	10
46. <i>Vespula germanica</i> (Fabricius, 1793)	Eusocial	Herbaceous plants	Spiders (Araneae) and other insects	omnivore	13

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Figure A1. Location of study regions in Hungary, in Europe and sampling sites.



Figure A2. Forest-steppe habitat photo from the Hungarian Great Plain region.